PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Programming Fundamentals Lab 08	**Instructor: Mr. Muhammad Hamza**
Program: AI - Fall 2025	**Date: 12th December, 2025**
Marks: 10	**Lab Engineer: Qandil Fatima**
Student Name:Eshal Ayub	**Reg. No:B25F1061AI100**

Instructions:

- Submit the file with your names following your registration numbers like **AI001_Name.**
- **Copy Paste work will be given no marks.**

Lab Task

1. Write a program that asks for the number of hours worked by six employees. It stores the values in an array.

```cpp
#include <iostream>
using namespace std;

int main() {
    int hours[6];

    cout << "Enter hours for 6 employees:\n";
    for(int i = 0; i < 6; i++) {
        cin >> hours[i];
    }

    cout << "\nHours entered:\n";
    for(int i = 0; i < 6; i++) {
        cout << "Employee " << i+1 << ": " << hours[i] << endl;
    }

    return 0;
}
```

```
9

2
```

Output:

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled3\cmake-build-debug\untitled3.exe"
Enter hours for 6 employees:
5

6

8

9

2

4


Hours entered:
Employee 1: 5
Employee 2: 6
Employee 3: 8
Employee 4: 9
Employee 5: 2
Employee 6: 4

Process finished with exit code 0
```

2. Write a program that asks the user to type 10 integers of an array. The program must compute and write how many integers are greater than or equal to 10.

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[10], count = 0;

    cout << "Enter 10 integers:\n";
    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
        if(arr[i] >= 10)
            count++;
    }

    cout << "\nNumbers >= 10: " << count;
    return 0;
}
```

Output:

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled3\cmake-build-debug\untitled3.exe"
Enter 10 integers:
1

2

3

4

5

6

7

8

9

0


Numbers >= 10: 0
Process finished with exit code 0
```

3. Write a C++ program to swap first and last element of an integer 1-d array

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[5];

    cout << "Enter 5 integers:\n";
    for(int i = 0; i < 5; i++)
        cin >> arr[i];

    int temp = arr[0];
    arr[0] = arr[4];
    arr[4] = temp;

    cout << "\nAfter swapping first and last:\n";
    for(int i = 0; i < 5; i++)
        cout << arr[i] << " ";

    return 0;
}
```

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled3\cmake-build-debug\untitled3.exe"
Enter 5 integers:
```

Output:

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled3\cmake-build-debug\untitled3.exe"
Enter 5 integers:
1
4
7
9
3

After swapping first and last:
3 4 7 9 1
Process finished with exit code 0
```

4. Write a C++ function that multiplies two matrices using arrays.

```cpp
#include <iostream>
using namespace std;

void multiply(int A[2][2], int B[2][2], int C[2][2]) {
    for(int i=0;i<2;i++) {
        for(int j=0;j<2;j++) {
            C[i][j] = 0;
            for(int k=0;k<2;k++)
                C[i][j] += A[i][k] * B[k][j];
        }
    }
}
int main() {
    int A[2][2], B[2][2], C[2][2];
    cout << "Enter Matrix A:\n";
    for(int i=0;i<2;i++) for(int j=0;j<2;j++) cin >> A[i][j];
    cout << "Enter Matrix B:\n";
    for(int i=0;i<2;i++) for(int j=0;j<2;j++) cin >> B[i][j];
    multiply(A, B, C);
    cout << "\nResult Matrix:\n";
    for(int i=0;i<2;i++) {
        for(int j=0;j<2;j++)
            cout << C[i][j] << " ";
        cout << endl;
    }
}
```

Output:

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled3\cmake-build-debug\untitled3.exe"
Enter Matrix A:
1

2

3

4

Enter Matrix B:
2

4

6

8


Result Matrix:
14 20
30 44

Process finished with exit code 0
```

5. Write a C++ function to add two matrices using multidimensional arrays.

```cpp
#include <iostream>
using namespace std;

int main() {
    int A[2][2], B[2][2], C[2][2];

    cout << "Enter Matrix A:\n";
    for(int i=0;i<2;i++) for(int j=0;j<2;j++) cin >> A[i][j];

    cout << "Enter Matrix B:\n";
    for(int i=0;i<2;i++) for(int j=0;j<2;j++) cin >> B[i][j];

    cout << "\nSum Matrix:\n";
    for(int i=0;i<2;i++) {
        for(int j=0;j<2;j++) {
            C[i][j] = A[i][j] + B[i][j];
            cout << C[i][j] << " ";
        }
        cout << endl;
    }

    return 0;
}
```

Output:

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled3\cmake-build-debug\untitled3.exe"
Enter Matrix A:
2

3

4

5

Enter Matrix B:
6

7

8

9


Sum Matrix:
8 10
12 14

Process finished with exit code 0
```

6. Write a program in C++ to find the transpose of a matrix.

```cpp
#include <iostream>
using namespace std;

int main() {
    int A[2][2], T[2][2];

    cout << "Enter 4 numbers for matrix:\n";
    for(int i=0;i<2;i++) for(int j=0;j<2;j++) cin >> A[i][j];

    for(int i=0;i<2;i++)
        for(int j=0;j<2;j++)
            T[j][i] = A[i][j];

    cout << "\nTranspose:\n";
    for(int i=0;i<2;i++) {
        for(int j=0;j<2;j++)
            cout << T[i][j] << " ";
        cout << endl;
    }

    return 0;
}
```

Output:

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled3\cmake-build-debug\untitled3.exe"
Enter 4 numbers for matrix:
5

2

4

6


Transpose:
5 4
2 6

Process finished with exit code 0
```

7. Write a function in C++ to find out the highest number in a 2 Dimensional array.

```cpp
#include <iostream>
using namespace std;

int main() {
    int A[2][2], maxVal;

    cout << "Enter 4 numbers:\n";
    for(int i=0;i<2;i++) for(int j=0;j<2;j++) cin >> A[i][j];

    maxVal = A[0][0];

    for(int i=0;i<2;i++)
        for(int j=0;j<2;j++)
            if(A[i][j] > maxVal)
                maxVal = A[i][j];

    cout << "\nHighest number = " << maxVal;
    return 0;
}
```

Output:

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled3\cmake-build-debug\untitled3.exe"
Enter 4 numbers:

2

4

9

6


Highest number = 9
Process finished with exit code 0
```

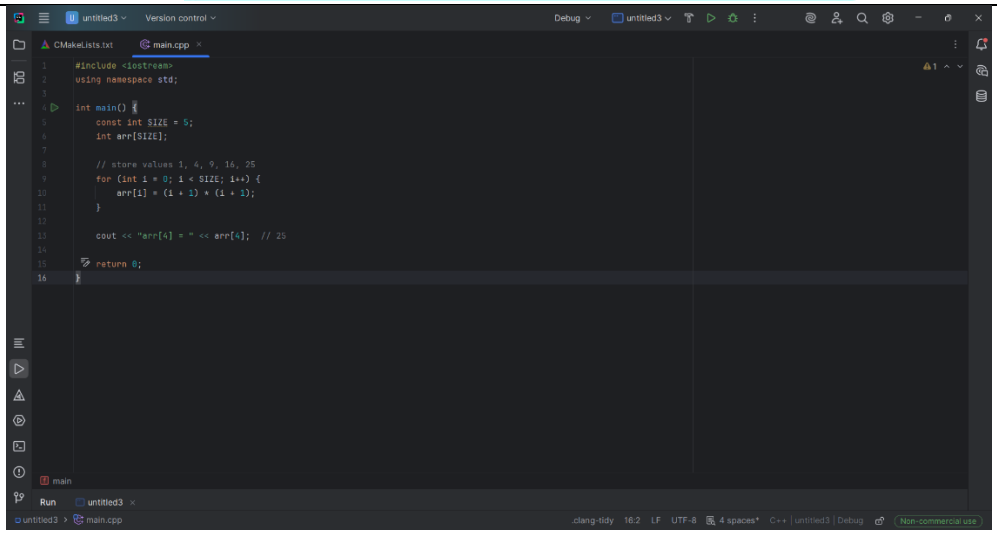
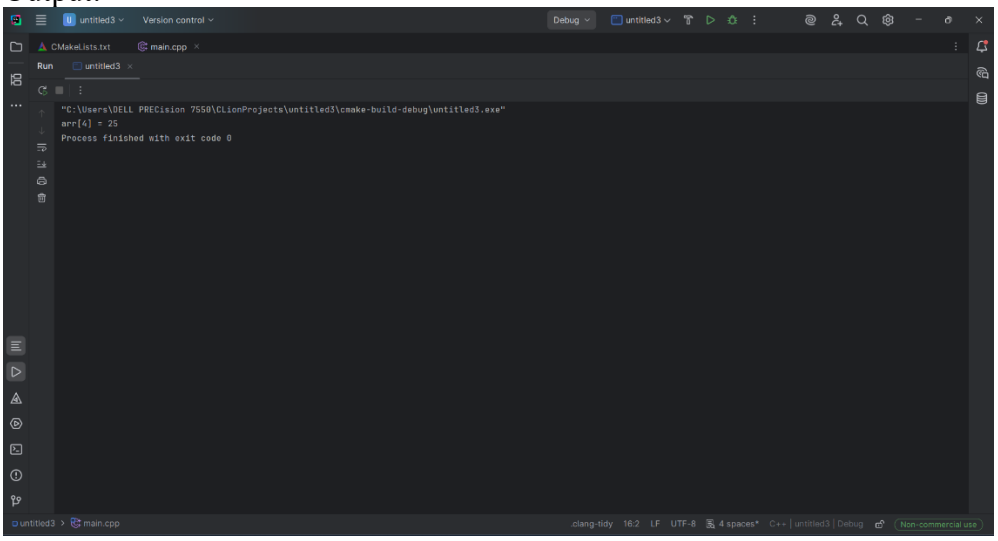
8. Write Correct the errors in the following program:

```
int main(){
    int SIZE=5;
    int arr[SIZE];

    // to store value 1 4 9 16 25 in arr
    for (int i=1;i<=5;i++)
         arr[i]=i*i;

    cout<<"arr5="<<arr[5];

    return 0;
}
```



Output: